Lab 4

DHCP

In this lab, for the first task, we will configure the DHCP server and set the DHCP-assigned network settings.
DHCP (Dynamic Host Configuration Protocol) is a client/server network protocol that is used to automatically assign IP addresses and other network configuration parameters to devices (clients) on a network. This eliminates the need for manual IP address configuration for each device, which simplifies network management. With DHCP, this entire process is automated and managed centrally. The DHCP server maintains a pool of IP addresses and leases an address to any DHCP-enabled client when it starts up on the network. Because the IP addresses are dynamic (leased) rather than static (permanently assigned), addresses no longer in use are automatically returned to the pool for reallocation.

How DHCP Works

The client device sends out a **Discover** broadcast message to the network to locate a DHCP server. If the DHCP server is not on the same network as the client, a DHCP relay agent on the network forwards the message to the DHCP server. The DHCP server, which is configured with a DHCP pool, receives the DHCP Discover message. The server selects an available IP address from its DHCP pool and prepares a DHCP **Offer** message that includes the assigned IP address, subnet mask, default gateway, DNS servers, and lease time. The client receives the offer and sends a **Request** to the selected DHCP server to accept the offered IP address. The DHCP server confirms the assignment and sends **Acknowledgement** to the client with the IP address, lease time, and other configuration details.

What is a DHCP pool?

A DHCP pool is a range of IP addresses that a DHCP server can allocate to clients on a network. When a device connects to the network, it requests an IP address from the DHCP server. The server then assigns an IP address from the pool to the device, allowing it to communicate on the network.

What is a DHCP relay?

DHCP relay, also known as DHCP forwarding, is a mechanism used to enable DHCP clients and servers to communicate across different networks. This is particularly useful when a DHCP server is not located on the same local network as the DHCP clients.

We can configure a DHCP server on a router using its CLI or a dedicated server.

DHCP Server Configuration and Management using a router

1. **Excluding Addresses**:
    - **Purpose**: Excluding addresses in a DHCP server prevents the server from assigning specific IP addresses to clients, ensuring those addresses are reserved for static devices or network services. This avoids IP conflicts and ensures reliable assignments for critical devices.
    - **Command**: ***ip dhcp excluded-address [Start IP address] [End IP address]***
2. **Creating a DHCP Pool**:
    - **Command**: ***ip dhcp pool [Pool name]***
3. **Configuring Pool Network Settings**:
    - **Network**: Defines the IP address range and subnet mask.
        - **Command**: ***network [Network address] [Subnet mask]***
    - **DNS Server**: Specifies the DNS server addresses for clients.
        - **Command**: ***dns-server [DNS address]***
    - **Default Router**: Sets the default gateway for clients.
        - **Command**: ***default-router [Default gateway]***
4. **Setting Up a DHCP Relay Agent**:
    - **Enter the Interface**: Access the interface to configure the relay agent.
        - **Command**: ***interface [Interface name]***
    - **Configure Relay Agent**: Specifies the IP address of the DHCP server to forward requests to.
        - **Command**: ***ip helper-address [DHCP server IP address]***
5. **Assigning an IP Address via DHCP**:
    - **Command**: ***ip address dhcp***
6. **Viewing DHCP Bindings**:
    - **Purpose**: Provides details about current DHCP leases, including allocated IP addresses, lease expiration times, and associated MAC addresses.
    - **Command**: ***show ip dhcp bindings***

ROUTING

As the first part of the lab, we will do some routing for the routers. Routing refers to determining the path between the source and the destination. Routers route a packet following a routing table. This routing table is going to be populated with the routing instances that will be configured by the user. Like most things, routing can be done both manually and dynamically. Manual routing is known as static routing, whereas dynamic routing just goes by its name.

Static Routing:

In the case of static routing, the administrator decides which path a packet should take upon arriving at the router. So, the administrator sets the exit option by him/herself. Static routing is done only for remote networks. The command for static routing is,

ip route [Target Network] [Target Network's Subnet Mask] [Exit Option]

Here [Target Network], and [Target Network's Subnet Mask] fields are pretty self-explanatory. We can write the [Exit Option] field in 3 ways.

1. Exit interface: We can directly put the interface name, through which the router will forward any packet for the target network. This type of routing is known as directly attached routing.
2. Next Hop IP: When the router forwards the packet, the packet will reach the immediate next router X. The interface of router X through which the packet will arrive at router X is known as the next hop. We will put the IP address of this interface as the next hop ip. This type of routing is known as recursive routing.
3. Fully specified: Here, we will provide both the exit interface and the next hop IP, which is why it is known as fully specified routing.

Types of Routing:

- Standard Static Routing: When we create a route only for a stub network, that is known as standard static routing. For standard static routing, the value of AD is by default 1. That's why we don't write the value of AD in the standard routing command.
- Default Static Routing: It is quite impossible for a router to know each and every network of the Internet. Also, it is quite troublesome to set static routing for all these networks. That's why each router has a default route, where it passes a packet if it's unaware of how to reach that destination. If there is a default route configured in a router, the router will initially check all the routing instances, and if it doesn't find any instance that matches the destination, it follows the last instance of the routing table, which is the default route. The command for default routing is,
  ip route 0.0.0.0 0.0.0.0 [Exit Option]

Try to find out why the values of the Target network and the target network's subnet mask are kept as such.

- Floating Static Routing: Sometimes, we need to create some backup routes in case the primary route is down. This type of routing is known as floating static routing. The command for floating static routing is
  ip route [Target Network] [Target Network's Subnet Mask] [Exit Option] [AD>1]
  You can see that the command is the same as other types of routing, except there is a new field called AD. If we don't set a value for AD, by default it will be 1. The lower the value, the more trustworthy the path is. For standard routing, the value is 1 by default; that's why we didn't write it. However, if we want to create a route as a backup, we make that route less trustworthy to the router. So, we set a value of AD, which is greater than 1. Also, we can create a backup route for both the standard route and the default route.

Dynamic Routing:

In the case of dynamic routing, the routers exchange information among themselves and get to know about the topology. The administrator just sets some rules/protocols for how the routers should communicate with each other. There are different types of dynamic routing protocols, and we will cover the Routing Information Protocol (RIP) version 2. In the case of RIP, each router shares the information of its directly connected networks with other routers. Details of this process will be discussed in class.